

DHCP SERVER CONFIGURATION

1. Check the Change Request

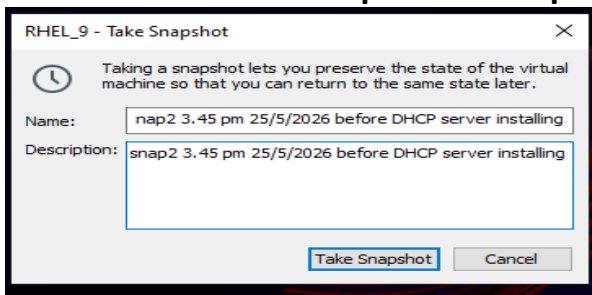
- Check whether the change is assigned to you by the TL or manager.
- Check the change status:
 - **Approved**
 - Pending
- If the change is approved, then start the work.

2. Access the Server

- Take the server IP from the TL.
- Login using:
 - **SSH**
 - RDP
 - VNC
 - Remote Console

3. Take Backup

- Always take backup before making changes.
- Backup types:
 - Physical Server → NetBackup (Full OS Backup)
 - **Virtual Machine → Snapshot Backup**



- AWS Instance → Snapshot Backup

4. Configure Network and Hostname

- **Network Adapter:** - bridged (Automatic)
- **Configure Static IP:-**

```
[root@dhcpsrv1 ~]# nmcli con mod ens160 ipv4.addresses 10.0.0.100/8 ipv4.gateway 10.0.0.1
[root@dhcpsrv1 ~]# nmcli con up ens160
```

```
[root@dhcpsrv1 ~]# ip a s ens160
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:0c:29:12:a4:79 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 10.0.0.100/8 brd 10.255.255.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 2409:40c2:100c:c702:20c:29ff:fe12:a479/64 scope global dynamic noprefixroute
        valid_lft 7037sec preferred_lft 7037sec
    inet6 fe80::20c:29ff:fe12:a479/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@dhcpsrv1 ~]#
```

- **Set the hostname:** -

```
[root@dhcpsrv1 ~]# hostnamectl set-hostname dhcpsrv1
```

```
[root@dhcpsrv1 ~]# hostname
dhcpsrv1
[root@dhcpsrv1 ~]#
```

- **Configure local DNS in /etc/hosts.**

```
[root@dhcpsrv1 ~]# vi /etc/hosts
```

```
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
10.0.0.100   dhcpsrv1      dhcpserver
```

- **Test IP, hostname, and internet connectivity.**

```
[root@dhcpsrv1 ~]# ping 10.0.0.100
PING 10.0.0.100 (10.0.0.100) 56(84) bytes of data.
64 bytes from 10.0.0.100: icmp_seq=1 ttl=64 time=0.164 ms
64 bytes from 10.0.0.100: icmp_seq=2 ttl=64 time=0.416 ms
^C
--- 10.0.0.100 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1010ms
rtt min/avg/max/mdev = 0.164/0.290/0.416/0.126 ms
[root@dhcpsrv1 ~]# ping dhcpsrv1
PING dhcpsrv1 (10.0.0.100) 56(84) bytes of data.
64 bytes from dhcpsrv1 (10.0.0.100): icmp_seq=1 ttl=64 time=0.108 ms
64 bytes from dhcpsrv1 (10.0.0.100): icmp_seq=2 ttl=64 time=0.099 ms
^C
--- dhcpsrv1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1023ms
rtt min/avg/max/mdev = 0.099/0.103/0.108/0.004 ms
```

5. Configure Repository

- **Local Repository**

```
[root@dhcpsrv1 ~]# mount /dev/sr0 /mnt
```

```
[root@dhcpsrv1 ~]# vi /etc/yum.repos.d/local.repo
```

```
[AppStreamrepo]
name=this is AppStream file base repo
baseurl=file:///mnt/AppStream
enabled=1
gpgcheck=0
```

```
[BaseOS]
name=this is BaseOS file base repo
baseurl=file:///mnt/BaseOS
enabled=1
gpgcheck=0
```

- RHN Repository
- EPEL Repository
- Third-party Repository
- Satellite Server Repository

6. Verify and Install Required Packages

- **Search required package.**

```
[root@dhcpsrv1 ~]# yum search dhcp*
```

```
dhcp-server.x86_64 : Provides the ISC DHCP server
```

- **Check information about package.**

```
[root@dhcpsrv1 ~]# yum info dhcp-server.x86_64
```

```
[root@dhcpsrv1 ~]# yum info dhcp-server.x86_64
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:02:07 ago on Mon 25 May 2026 06:08:09 PM IST.
Installed Packages
Name       : dhcp-server
Epoch     : 12
Version    : 4.4.2
Release    : 19.b1.el9
Architecture : x86_64
Size       : 3.9 M
Source     : dhcp-4.4.2-19.b1.el9.src.rpm
Repository : @System
From repo  : BaseOS
Summary    : Provides the ISC DHCP server
JURL       : https://www.isc.org/dhcp/
License    : ISC
Description: DHCP (Dynamic Host Configuration Protocol) is a protocol which allows
            : individual devices on an IP network to get their own network
            : configuration information (IP address, subnetmask, broadcast address,
            : etc.) from a DHCP server. The overall purpose of DHCP is to make it
            : easier to administer a large network.
            :
            : This package provides the ISC DHCP server.
```

- **Install the required package with the correct version.**

```
[root@dhcpsrv1 ~]# yum install -y dhcp-server.x86_64
```

```
Installed:
  dhcp-common-12:4.4.2-19.b1.el9.noarch      dhcp-server-12:4.4.2-19.b1.el9.x86_64

Complete!
```

- **Check installed packages.**

```
[root@dhcpsrv1 ~]# rpm -qa dhcp-server
dhcp-server-4.4.2-19.b1.el9.x86_64
```

7. Configure the Server

- **Find the configuration file.**

```
[root@dhcpsrv1 ~]# rpm -qa dhcp-server
```

```
[root@dhcpsrv1 ~]# rpm -qa dhcp-server
dhcp-server-4.4.2-19.b1.el9.x86_64
[root@dhcpsrv1 ~]# ^C
[root@dhcpsrv1 ~]# rpm -qc dhcp-server
/etc/dhcp/dhcpd.conf
/etc/dhcp/dhcpd6.conf
/etc/openssl/schema/dhcp.schema
/etc/sysconfig/dhcpd
/var/lib/dhcpd/dhcpd.leases
/var/lib/dhcpd/dhcpd6.leases
[root@dhcpsrv1 ~]# █
```

- **Take backup of the configuration file.**

```
[root@dhcpsrv1 ~]# cp /etc/dhcp/dhcpd.conf /tmp/dhcpd.conf.bkp
```

- **We use Template for configuration :**

- Change document

- Official documentation if needed

- **Template:** - see /usr/share/doc/dhcp-server/dhcpd.conf.example

```
[root@dhcpsrv1 ~]# cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcpd.conf
```

- **Configure this file:** -

```
[root@dhcpsrv1 ~]# vi /etc/dhcp/dhcpd.conf
```

```
#####created for dhcp serv
```

```
default-lease-time 600;  
max-lease-time 7200;
```

```
log-facility local7;
```

```
subnet 10.0.0.0 netmask 255.0.0.0 {  
    range 10.0.0.10 10.0.0.30;  
    #Default gateway  
    option routers 10.0.0.2;  
}
```

```
#host fantasia {  
    # hardware ethernet 08:00:07:26:c0:a5;  
    #fixed-address fantasia.example.com;  
#}  
"/etc/dhcp/dhcpd.conf" 124L, 3633B
```

8. Local Testing

- Check configuration syntax.

```
[root@dhcpsrv1 ~]# dhcpd -t
```

```
Internet Systems Consortium DHCP Server 4.4.2b1  
Copyright 2004-2019 Internet Systems Consortium.  
All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
Warning: subnet 10.0.0.0/8 overlaps subnet 10.5.5.0/27  
ldap_gssapi_principal is not set,GSSAPI Authentication for LDAP will not be used  
Not searching LDAP since ldap-server, ldap-port and ldap-base-dn were not specified in the config file  
Config file: /etc/dhcp/dhcpd.conf  
Database file: /var/lib/dhcpd/dhcpd.leases  
PID file: /var/run/dhcpd.pid  
Source compiled to use binary-leases
```

- Start, stop, and check service status.

```
# systemctl start dhcpd
```

```
# systemctl status dhcpd
```

```
[root@dhcpsrv1 ~]# systemctl start dhcpd  
[root@dhcpsrv1 ~]# systemctl status dhcpd  
● dhcpd.service - DHCPv4 Server Daemon  
   Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; disabled; preset: disabled)  
   Active: active (running) since Mon 2026-05-25 18:51:45 IST; 17s ago  
     Docs: man:dhcpd(8)  
           man:dhcpd.conf(5)  
  Main PID: 4032 (dhcpd)  
   Status: "Dispatching packets..."  
    Tasks: 1 (limit: 10726)  
  Memory: 4.7M  
    CPU: 29ms  
   CGroup: /system.slice/dhcpd.service  
           └─4032 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid
```

```
May 25 18:51:44 dhcpsrv1 dhcpd[4032]: Copyright 2004-2019 Internet Systems Consortium.  
May 25 18:51:44 dhcpsrv1 dhcpd[4032]: All rights reserved.  
May 25 18:51:44 dhcpsrv1 dhcpd[4032]: For info, please visit https://www.isc.org/software/dhcp/  
May 25 18:51:44 dhcpsrv1 dhcpd[4032]: Source compiled to use binary-leases  
May 25 18:51:44 dhcpsrv1 dhcpd[4032]: Wrote 0 leases to leases file.  
May 25 18:51:45 dhcpsrv1 dhcpd[4032]: Listening on LPF/ens160/00:0c:29:12:a4:79/10.0.0.0/8  
May 25 18:51:45 dhcpsrv1 dhcpd[4032]: Sending on LPF/ens160/00:0c:29:12:a4:79/10.0.0.0/8  
May 25 18:51:45 dhcpsrv1 dhcpd[4032]: Sending on Socket/fallback/fallback-net  
May 25 18:51:45 dhcpsrv1 dhcpd[4032]: Server starting service.  
May 25 18:51:45 dhcpsrv1 systemd[1]: Started DHCPv4 Server Daemon.
```

```
# systemctl stop dhcpd
```

- Check logs:-

```
May 25 18:51:45 dhcpsrv1 systemd[1]: Started DHCPv4 Server Daemon.  
Subject: A start job for unit dhcpd.service has finished successfully  
A start job for unit dhcpd.service has finished successfully.  
May 25 18:52:13 dhcpsrv1 dhcpd[4032]: DHCPREQUEST for 172.18.126.202 (172.18.126.223) from 00:0c:29:5b:  
:60 via ens160: unknown lease 172.18.126.202.  
May 25 18:52:37 dhcpsrv1 systemd[1]: Stopping DHCPv4 Server Daemon...  
Subject: A stop job for unit dhcpd.service has begun execution  
A stop job for unit dhcpd.service has begun execution.  
May 25 18:52:37 dhcpsrv1 systemd[1]: dhcpd.service: Deactivated successfully.  
The unit dhcpd.service has successfully entered the 'dead' state.  
May 25 18:52:37 dhcpsrv1 systemd[1]: Stopped DHCPv4 Server Daemon.  
Subject: A stop job for unit dhcpd.service has finished  
A stop job for unit dhcpd.service has finished.
```

- Check port numbers.

```
[root@dhcpsrv1 ~]# ss -antlpu | grep dhcp
```

```
[root@dnpcsrvt ~]# systemctl start dnpcd
[root@dhcpcsrvt ~]# ss -antlpu | grep dhcp
udp UNCONN 0 0 0.0.0.0:67 0.0.0.0:* users:(("dhcpd",pid=4104,fd=7))
```

If testing fails, go back and reconfigure the settings.

9. Enable Service Permanently

```
[root@dhcpcsrvt ~]# systemctl enable --now dhcpd
```

```
[root@dhcpcsrvt ~]# systemctl status dhcpd
● dhcpd.service - DHCPv4 Server Daemon
   Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-05-25 18:54:49 IST; 3min 14s ago
     Docs: man:dhcpd(8)
           man:dhcpd.conf(5)
  Main PID: 4104 (dhcpd)
    Status: "Dispatching packets..."
     Tasks: 1 (limit: 10726)
  Memory: 4.7M
     CPU: 31ms
   CGroup: /system.slice/dhcpd.service
           └─4104 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid

May 25 18:54:49 dhcpcsrvt dhcpd[4104]: Source compiled to use binary-leases
May 25 18:54:49 dhcpcsrvt dhcpd[4104]: Wrote 0 leases to leases file.
May 25 18:54:49 dhcpcsrvt dhcpd[4104]: Listening on LPF/ens160/00:0c:29:12:a4:79/10.0.0.0/8
May 25 18:54:49 dhcpcsrvt dhcpd[4104]: Sending on LPF/ens160/00:0c:29:12:a4:79/10.0.0.0/8
May 25 18:54:49 dhcpcsrvt dhcpd[4104]: Sending on Socket/fallback/fallback-net
```

10. Temporarily disable Firewall for remote checking

```
[root@dhcpcsrvt ~]# systemctl stop firewalld.service
```

11. Temporarily disable SELinux for remote checking

```
[root@dhcpcsrvt ~]# setenforce 0
```

12. Test from remote machine

- **Network Adopter:** - bridged (Automatic)
- **Restart Client Network**
[root@localhost Desktop]# nmcli con up ens160
- **Verify IP Address**

```
[root@localhost Desktop]# ip a s ens160
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:5b:da:60 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.10/8 brd 10.255.255.255 scope global dynamic noprefixroute ens160
        valid_lft 382sec preferred_lft 382sec
    inet6 fe80::20c:29ff:fe5b:da60/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@localhost Desktop]#
```

- **Test Connectivity**

```
[root@localhost Desktop]# ping 10.0.0.100
```

```
[root@localhost Desktop]# ping 10.0.0.100
PING 10.0.0.100 (10.0.0.100) 56(84) bytes of data.
64 bytes from 10.0.0.100: icmp_seq=1 ttl=64 time=1.95 ms
64 bytes from 10.0.0.100: icmp_seq=2 ttl=64 time=0.950 ms
^C
--- 10.0.0.100 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.950/1.448/1.947/0.499 ms
[root@localhost Desktop]#
```

Final Verification

- DHCP service is running successfully.
- Client machine receives IP automatically.
- Network connectivity is working properly.
- Re-enable Firewall and SELinux after successful testing.